RAJALAKSHMI ENGINEERING COLLEGE

Lathikka D 251001133-2025-IT

L2

Started on	Friday, 4 September 2026, 9:50 PM
State	Finished
Completed on	Friday, 4 September 2026, 9:51 PM
Time taken	43 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```
#include <stdio.h>

int majorityElement(int arr[], int low, int high)
{
    if (low == high)
        return arr[low];

    int mid = (low + high) / 2;

    int left = majorityElement(arr, low, mid);
    int right = majorityElement(arr, mid + 1, high);

    if (left == right)
        return left;

    int leftCount = 0, rightCount = 0;

    for (int i = low; i <= high; i++)
    {
        if (arr[i] == left)
            leftCount++;

        if (arr[i] == right)
            rightCount++;
    }

    if (leftCount > rightCount)
        return left;
    else
        return right;
}

int main()
{
```

```
35      int n;
36      scanf("%d", &n);
37
38      int nums[n];
39
40      for (int i = 0; i < n; i++)
41          scanf("%d", &nums[i]);
42
43      printf("%d\n", majorityElement(nums, 0, n - 1));
44
45      return 0;
46  }
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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